

AP CSP | Lesson 29: Privacy & Security

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: C

C includes descriptive and linkable information and the effect of combining data. A excludes many forms of PII. B incorrectly requires unique identification by one field. D ignores location, descriptive fields and links to other data.

Q2. Answer: A

A supplies the group and measure needed for the stated summary. B omits the rating and adds unnecessary identifying detail. C expands collection without a stated purpose. D collects sensitive authentication information unrelated to the summary.

Q3. Answer: D

D combines something known with verified possession. A, B and C use only the knowledge category, even when there are two secrets or two steps.

Q4. Answer: B

B uses the recipient's matched keys in the confidentiality model. A assigns the sender's keys to the wrong purpose. C reverses roles and exposes the secret. D ignores that the public and private keys must be the matching pair.

Q5. Answer: B and C

B is a reason to keep software updated; it is not a promise of perfect security. C explains why familiar names alone are insufficient verification. A overstates certificates, which validate key ownership through a trust model. D confuses public access with unauthorized access points.

Q6. Answer: C

C directly names recording keystrokes. A combines records into summaries. B transforms data to protect confidentiality using matched keys. D reduces representation size; none of those definitions describes capturing typed credentials.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Use zone to group reports and shadeRating to summarize need. Omit name, email and exactGPS because this purpose does not require identifying, contacting or precisely locating a respondent. Aggregation reduces detail, but small groups or combination with other data can still permit inferences about people.

1 mark: Identifies zone and explains that it groups responses into North/South.

1 mark: Identifies shadeRating and explains that it supplies the measure of shade need.

1 mark: Identifies name, email and exactGPS as unnecessary for this stated aggregate purpose, with a purpose-based reason.

1 mark: Explains a plausible remaining privacy risk from small groups or linking summaries to other data; does not claim zero risk.

Q8. 4 marks

Answer: A password and a PIN are both knowledge factors. Verified phone approval is possession. Password plus verified phone approval therefore combines two categories; password plus PIN does not. MFA reduces account-access risk, but does not guarantee protection against every phishing attempt, malware or misuse by an authorized user.

1 mark: Classifies password and PIN as the knowledge category.

1 mark: Classifies verified phone approval as the possession category.

1 mark: Identifies password plus phone as MFA and password plus PIN as one category.

1 mark: Explains one plausible remaining risk, such as malware, deceptive approval requests or misuse after access; does not say MFA guarantees safety.

Q9. 4 marks

Answer: Encrypt with Blake's public key. Decrypt with Blake's private key. Blake can share the public key while protecting the private key. Symmetric encryption instead uses the same secret key for encryption and decryption, so authorized parties must share and protect that secret.

1 mark: Identifies Blake's public key for encryption.

1 mark: Identifies Blake's private key for decryption.

1 mark: Explains that the public key may be shared while the private key must remain secret.

1 mark: Explains that symmetric encryption uses the same secret key at both ends, unlike the public/private pair.

Q10. 4 marks

Answer: The form may be phishing that attempts to obtain credentials; verify using an independently known channel and do not provide the password through the message. The untrusted download may install malware, including a keylogger; obtain software or updates from a trusted source instead. A familiar account may be compromised.

1 mark: Identifies the credential-request message as a phishing risk, including why the request is suspicious.

1 mark: Gives a safe independent verification or reporting response and avoids providing credentials through the supplied form.

1 mark: Identifies the untrusted download as a malware or keylogging risk.

1 mark: Gives a matched protective action such as using a trusted update source and keeping verified software updated; does not claim any download is automatically safe.

Q11. 4 marks

Answer: Encryption protects the confidentiality of data in transit. A certificate authority uses a trust model to validate ownership of encryption keys through digital certificates. Neither proves that a request is honest or appropriate. Do not reveal the password in response; verify the request through an independently known service or contact channel.

1 mark: Explains that encryption protects confidentiality during transfer, assuming the intended key association.

1 mark: Explains that a certificate authority validates encryption-key ownership using digital certificates and a trust model.

1 mark: Explains that channel encryption or a valid certificate does not prove message honesty or eliminate phishing.

1 mark: Gives a safer independent verification response that does not disclose the password to the request.

Q12. 4 marks

Answer: A rogue access point is an unauthorized wireless access point that can give unauthorized access to a secure network. Public availability does not by itself make a wireless network unauthorized or malicious. Reviewing and removing unnecessary app permissions limits data or device access. Software updates can repair known security flaws, although no system becomes risk-free.

1 mark: Defines a rogue access point as unauthorized wireless access that can enable unauthorized access to a secure network.

1 mark: Explains that public availability alone does not mean the access point is unauthorized or malicious.

1 mark: Explains that reviewing or removing unnecessary permissions reduces unnecessary app access to data or functions.

1 mark: Explains that updates can patch known flaws without guaranteeing perfect protection.